

Figure 1: Successful Remote Exploitation using metasploit.

F002 – SQL Injection

Tool Used: Manual Testing (DVWA)

Severity: High

Description: User data retrieved using SQL injection payload.

Impact: Database exposure risk.

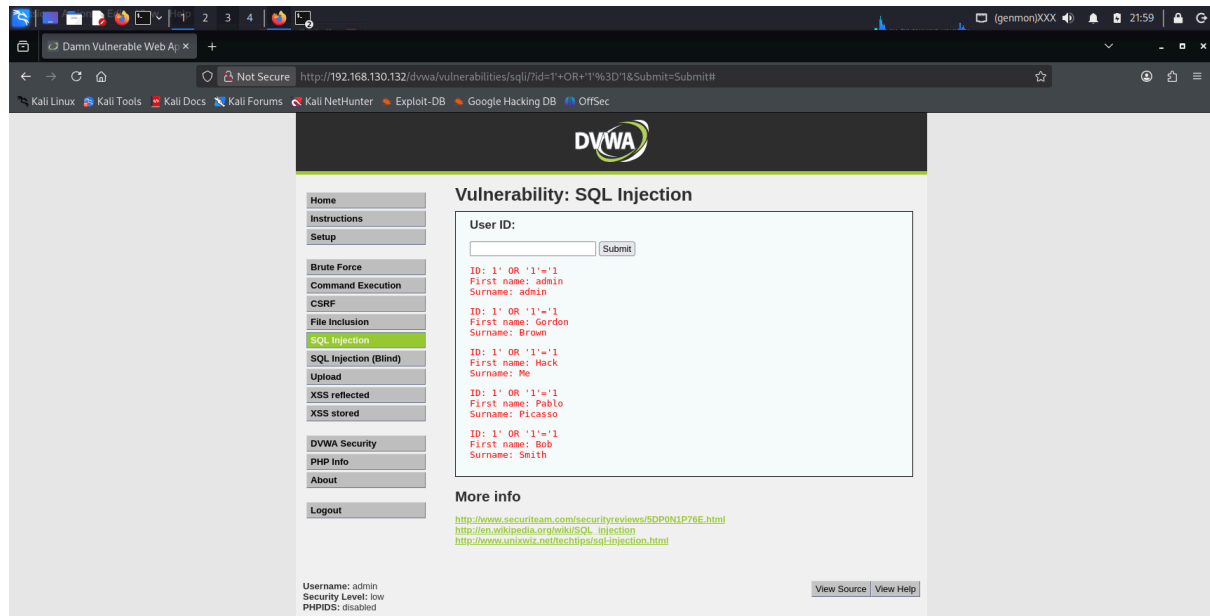


Figure 2: SQL Injection vulnerability demonstration.

F003 – Reflected XSS

Tool Used: Manual Testing (DVWA)

Severity: Medium

Description: JavaScript alert executed via input field.

Impact: Session hijacking / data theft possible.

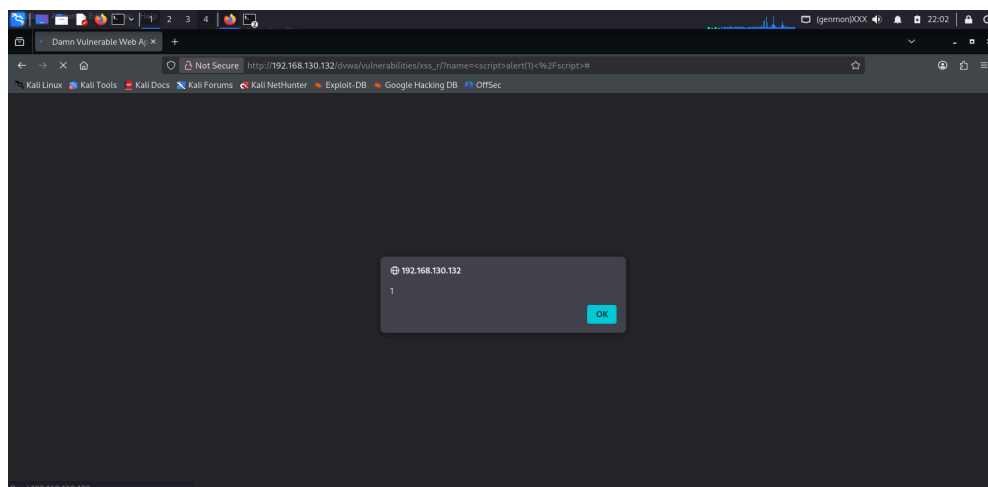


Figure 3: Reflected XSS alert popup.

3. Findings Table

Finding ID	Vulnerability	Severity	Remediation
F001	Remote Exploit	Critical	Update server, close unused ports
F002	SQL Injection	High	Use prepared statements
F003	Reflected XSS	Medium	Input sanitization & output encoding

4. Remediation

- Implement input validation and sanitization
- Use prepared SQL statements
- Apply output encoding for user inputs
- Keep system and services updated

5. Evidence Summary

The penetration testing exercise successfully demonstrated multiple vulnerabilities within the testing environment. Evidence was collected through screenshots of exploitation results, SQL injection outputs, and XSS alert execution. These findings confirm the presence of critical and high-risk vulnerabilities requiring immediate remediation.

6. Conclusion

The assessment revealed critical security gaps in the web application and system configuration. Implementing recommended remediation steps will significantly improve the overall security posture and reduce the risk of exploitation.